Max’s Bots

Lily Urbanczyk | lurbancz@asu.edu

Kolter Nelson | knels104@asu.edu

Edward Bernstein | egbernst@asu.edu

Carlos Martinez | camar105@asu.edu

Oscar Barrera | oabarre2@asu.edu

Carter Baldwin | cdbaldw4@asu.edu

MAE 301 Applied Experimental Statistics

Max Ren

Arizona State University

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1. Problem Statement

Who is the user?

Undergraduate students enrolled in introductory statistics courses. Particularly college students with no strong statistics math background who are taking the course as a requirement. Secondarily, the professor who wants their students to have a study tool in the course, offering them consistent course-accurate support outside of office hours.

What problem do they experience today?

Students already have access to powerful AI tools like ChatGPT, Claude, and Gemini, but these tools have an underlying problem; these AI tools do not know their user. These AIs typically explain expecting the user to understand course related jargon, concepts throughout the entirety of the course, or have substantial mathematical experience. Furthermore, using AI tools can take more time for the user who is clearly confused. A student has to consistently mention to AI, “don’t use jargon only a statistician would know”, “only cover material in a statistics course at week 5”, “remember I struggle with central limit theorem”. Most students don’t remember to mention all these things, or at least don’t want to constantly repeat themselves. As a result, AI can bring correct and immediate solutions, but a main problem with AI that is often looked past is its ability to help you understand, save time, and trust the information it delivers based on the resources it's using.

2. Why Now?

Why does this problem matter in the next 3-5 years?

AI study tools are rapidly becoming a standard part of higher education. Within 3 to 5 years, every student will have experience accessing AI whether it be for personal use, a study tool, or as an integration to their course work. AI will be transformed into a tool to help workflow, but this can make or break a student's learning experience. The students who understand the AI system and know how to correctly prompt AI will have an advantage over those who don't. This can create a learning gap, and discourage students. A tool that removes the need to prompt correctly can level the playing field in college use.

What changed (technology, regulations, culture) that makes this possible now?

There are three fundamental changes in technology that makes this possible now. Retrieval-Augmented Generation became accessible to developers and removed the need for specific research backgrounds which makes it practical to build an AI that can use resources from a larger source rather than specific training given to it by its developer. Additionally, AI applications, like OpenAI, are cheap enough to make it possible to create an economically personalized AI learning tool without the need for institutional funding. Lastly, the target consumer is not hard to reach or convince into using the AI tool as AI is already a widely and routinely used application to help students in college.

3. Proposed AI-Powered Solution

What does your product do for the user?

StatAI-TA is our personalized AI tool on track with the class's timing, structure, and stores memory of the student's progress resourced from the web and professor uploads of their own statistics course materials (syllabus, lecture slides, notes, textbook chapters, and recorded lectures) and interacts with students using three simple commands, "daily quiz", "summary", and "explain topic". Daily quiz prompts our AI to generate 10 questions from the professor's given course materials and weighs toward topics the student has struggled with in past lectures. Summary produces a plain language recap of any covered material and topics using the professor's own course material as the source. Explain topic, gives a concise targeted explanation dependent on the professor's uploaded resources and written without statistical jargon.

Where does AI/ML add unique value vs simple rules / heuristics?

AI adds unique value in this application by generating quiz questions specifically from the given resources and course content, adapts explanations and quizzes based on the student's demonstrated knowledge, and follows the same resources as the student, avoiding potential areas of spreading misinformation or assumptions that the student's knowledge is more advanced.

4. Initial Technical Concept

What data would you need (or already have)?

- Professor (or student)- uploaded course materials in pdf format including syllabus, lecture slides, textbook excerpts, personal notes, and recorded lecture.
- Student quiz and chat response history which is stored locally to track weak spots.
- No external datasets are required, the system is powered by the professor's (or student's) uploaded course material.

What model(s) might you use (e.g., GPT-style text model, classifier, recommender, vision model)?

- A GPT-style language model can allow explanations to be generated, quizzes to be created, and create summaries that are easy to understand
- An embedding model would help the system search uploaded course materials and retrieve the most relevant lecture slides, notes, or syllabus sections before answering.
- A simple classifier or progress-tracking model could be used to identify which statistics topics a student struggles with most based on quiz results.
- A vision-enabled model may also be useful for reading diagrams, charts, and image-based content in PDF lecture slides.

How could your nanoGPT work feed into this (e.g., custom generative component, fine-tuning, evaluation)

- nanoGPT work can be used in this project as a way to test or experiment with the language used to explain the problems generated. Using a small set of questions, we can use nanoGPT work as a guideline for future examples and notice patterns that are commonly repeated when explaining a problem. This will allow us to understand the

common strengths and weaknesses of generative models and allow us to work around them to better fine-tune future versions of the model.

5. Scope for MVP

What can you realistically build in ~6 weeks?

Since we are not skilled in coding, we may not be able to create a large-scale application where it can be accessed by a large classroom and a shared course AI but we can realistically build an individual AI tool that does all these things except the student has to input all the course material. Essentially, we can create a web app where one student uploads their own statistics course materials and uses the three commands as discusses to receive personalized course-grounded responses. The result should be the same as the professor uploading their own material. There will be no class-wide sharing and no persistent database but quiz history will be tracked.

Define a very concrete v1 feature: "A user can ___ and our system returns ___."

A user can upload their statistics course syllabus and lecture slides as PDFs, type "daily quiz", "summary", "explain" into a chat interface and our system returns a direct concise explanation or sets of 10 quiz questions and stores information about the student's knowledge level.

6. Risks and Questions

Top 3 Unknowns (e.g., data availability, evaluation, user adoption).

1. AI's ability to read the PDFs uploaded

Lecture slides exported as PDFs often contain images, diagrams, and certain formatting that AIs struggle with extracting and can lead to incomplete or inaccurate readings if not clear. Thus the AI's responses can become inaccurate based on bad input. We need to test how AI can adapt to these potential issues with real statistics course slides to know at what level AI cannot accurately read incomplete information.

2. Keeping the AI on topic

Language models can drift outside of the given content since they have access to the web and might generate a response with information that can defeat the purpose of staying within the timeline of the course and sticking to the professor's material if it feels that it is necessary to retrieve more information to answer a question. Especially for students who don't know any better and cannot verify statistical claims, it can offer a useless solution. It is important that we test this by prompting the AI and understanding where we can possibly demand the AI to avoid retrieving external resources.

3. Implementing "plain language" is subjective

An important factor of our application is that we want our explanations to avoid advanced statistical jargon. But how can we measure this? There is no metric for helping the AI stick to a range of suitable words that will help students unless we are extremely specific, yet AI can still

find subjectivity in this and use other jargon that the student may not be familiar with. User testing on real students is the only reliable evaluation.

7. Planned Data Sources

Our planned data sources are student-uploaded PDFs, knowledge base for RAG pipeline including syllabus, slides, and notes, OpenAI API, response generation for all three commands, converting document chunks to vectors, chromaDB, local vector store for document search, quiz generation.

Parts for the video

1. Introduction: Students actually don't get the most out of AI, it might be harming them more than benefiting them. (Explain how) **Lily**
2. What the solution is to this issue (our app) and how this works. **Carter Baldwin**
3. What models and tools will we use to make the app possible (What AI?) **Kolter**✔️
4. What can we realistically build & Risks and Issues (Closing) **Carlos Martinez**